

MARSABIT, Kenya

WMO: 636410

Lat: 2.300N

Lon: 37.900E

Elev: 1345

StdP: 86.18

Time Zone: 3.00 (E03)

Period: 90-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 14.2	(c) 14.8	(d) 4.6	(e) 6.2	(f) 23.0	(g) 7.2	(h) 7.4	(i) 23.1	(j) 25.0	(k) 17.0	(l) 21.3	(m) 17.1	(n) 11.9	(o) 160	(p) 0.917

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 2	(b) 7.8	(c) 27.8	(d) 17.7	(e) 27.1	(f) 17.6	(g) 26.4	(h) 17.5	(i) 20.2	(j) 23.8	(k) 19.8	(l) 23.3	(m) 19.5	(n) 23.0	(o) 6.8	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	16.4	21.3	18.8	16.1	21.0	18.5	15.7	20.8	64.8	23.9	63.3	23.4	62.1	23.3	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.1	(b) 19.2	(c) 17.8	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 21.1	(b) 19.2	(c) 17.8	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.8	21.8	22.5	22.3	21.5	21.2	20.0	19.0	19.2	20.2	21.0	20.5	20.7		
	DBStd	1.60	1.44	1.38	1.34	1.31	0.91	0.97	1.14	1.18	1.10	1.30	1.14	1.14		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	13	1	0	0	0	0	1	6	4	1	1	1	1		
	CDD10.0	3950	365	351	381	344	347	300	280	287	307	340	316	333		
	CDD18.3	922	108	117	123	94	88	51	28	32	57	82	67	75		
	CDH23.3	2217	294	497	483	229	137	63	27	50	119	187	55	76		
	CDH26.7	106	7	39	33	13	2	1	1	3	5	0	0	1		
(14) Wind	WSAvg	10.1	8.7	10.1	9.9	9.4	10.4	10.4	10.9	11.0	11.2	10.6	8.9	9.6		
(15) Precipitation	PrecAvg	682	33	29	45	177	71	11	8	10	3	74	113	71		
	PrecMax	1245	277	230	121	382	245	69	27	40	12	331	415	317		
	PrecMin	126	0	0	0	5	0	0	0	0	0	0	15	0		
	PrecStd	282	63	57	40	111	74	15	8	11	3	81	89	79		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.6	28.9	28.7	28.1	26.6	26.2	25.1	26.1	26.8	27.1	26.0	26.7		
		MCWB	16.6	17.1	17.8	18.2	18.6	17.9	17.0	17.4	17.3	17.9	18.1	17.0		
	2%	DB	26.7	27.7	27.7	26.8	25.5	24.9	24.0	24.4	25.4	26.0	24.8	25.4		
		MCWB	16.9	17.2	18.0	18.5	18.4	17.4	16.6	16.5	16.7	17.6	18.5	17.3		
	5%	DB	25.8	26.9	26.9	25.7	24.8	24.1	23.3	23.7	24.7	25.2	24.0	24.4		
		MCWB	16.9	17.1	17.8	18.7	18.5	16.9	16.3	16.2	16.4	17.4	18.5	17.4		
	10%	DB	25.1	26.2	26.0	24.7	24.1	23.3	22.6	23.0	23.9	24.4	23.2	23.5		
		MCWB	16.8	17.0	17.8	18.7	18.4	16.7	16.1	15.9	16.2	17.2	18.4	17.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.7	20.0	20.6	20.7	20.6	18.9	18.8	18.3	18.4	19.7	20.2	19.9		
		MCDB	23.7	24.9	24.6	24.0	24.2	23.8	23.4	23.8	24.7	22.9	23.2	22.9		
	2%	WB	19.2	19.2	19.8	20.2	19.9	18.2	17.6	17.5	17.7	19.1	19.6	19.5		
		MCDB	23.3	24.3	23.9	23.5	23.5	23.3	22.6	22.8	23.5	22.1	22.6	22.4		
	5%	WB	18.7	18.7	19.4	19.7	19.4	17.7	17.0	16.8	17.2	18.7	19.2	19.1		
		MCDB	22.9	24.0	23.6	23.0	23.0	22.7	21.8	22.2	22.9	21.8	22.1	21.9		
	10%	WB	18.4	18.3	19.0	19.4	19.0	17.3	16.5	16.4	16.7	18.3	18.9	18.7		
		MCDB	22.5	23.6	23.3	22.6	22.5	22.0	21.3	21.7	22.3	21.6	21.7	21.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	7.8	7.5	6.4	6.7	7.6	7.8	8.3	8.7	7.3	5.5	5.4		
		MCDBR	7.5	8.6	8.8	8.1	7.3	8.3	8.4	9.1	9.2	8.5	6.6	6.9		
	5% WB	MCWBR	3.6	3.2	2.9	2.8	2.5	2.9	2.8	3.3	2.9	2.7	2.4	2.8		
		MCWBR	6.5	7.7	7.1	6.2	6.4	7.9	7.9	8.6	8.9	6.7	5.5	5.2		
(40) Clear-Sky Solar Irradiance	taub	taub	0.342	0.362	0.372	0.358	0.355	0.412	0.415	0.375	0.353	0.347	0.356	0.349		
		taud	2.515	2.447	2.437	2.525	2.533	2.307	2.287	2.411	2.479	2.539	2.514	2.516		
	Ebn at Noon	Ebn at Noon	984	966	950	944	925	859	861	915	954	969	962	971		
		Edh at Noon	109	119	120	108	103	127	131	119	114	107	109	108		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	6.48	6.74	6.50	5.87	6.15	5.85	5.77	6.19	6.67	6.09	5.50	5.88		
	RadStd	RadStd	0.33	0.23	0.41	0.27	0.17	0.24	0.25	0.19	0.17	0.24	0.24	0.38		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page